

*?? **Did You Know?** Two Sigma, a \$60 billion quantitative hedge fund, spends more time designing the data structures for their trading systems than writing the actual trading algorithms. According to their engineers, well-organized data makes efficient algorithms possible, while disorganized data makes even the best algorithms fail. They have a saying internally: "Months of algorithm development can save you hours of proper data structure design." This insight—that data organization precedes successful analysis—is a key lesson from one of the world's most successful quantitative trading firms.*

Project 5: Multi-Asset Portfolio Management System

Objective: Create a comprehensive portfolio management system that uses different Python data structures to organize, analyze, and report on a multi-asset investment portfolio.

Requirements:

1. Create a portfolio using nested dictionaries with at least 8 securities across different asset classes (stocks, bonds, ETFs, etc.) including:
 - For each security: ticker symbol, purchase price, current price, quantity, purchase date, asset class, and sector
 - At least 3 different asset classes and 4 different sectors